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AQA GCSE Geography (8035) — Paper 2: Challenges in the Human Environment · 3.2.3 The Challenge of Resource Management (option: food)

2.5 Food

1. Areas of Food Surplus and Food Deficit

Food security — when all people, at all times, have access to enough safe and nutritious food to lead an active and healthy life. **Food insecurity** — when people lack reliable access to enough affordable, nutritious food.

Calorie intake — the number of calories a person consumes per day. An adult needs roughly 2,000–2,500 calories a day.

- **Food surplus** (producing more than is needed): North America, Europe, Australia, Argentina, Brazil. These areas have fertile soils, suitable climates, mechanised farming and the wealth to import food.
- **Food deficit**: much of sub-Saharan Africa, parts of South Asia, Yemen, Afghanistan and Haiti, where average calorie intake can be below 2,200 a day.

2. Reasons for Increasing Global Food Consumption

- **Population growth** — the world population passed 8 billion in 2022 and may reach nearly 10 billion by 2050.
- **Economic development** — as people in NEEs such as China and India become wealthier, they eat more food, especially meat and dairy, which take far more land, water and grain to produce.
- **Changing diets** — demand for processed food and year-round exotic produce.
- **Biofuels** — crops such as maize and sugar cane grown for fuel instead of food.

3. Factors Affecting Food Supply

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Factor	Effect on food supply
Climate	Crops need suitable temperatures and reliable rainfall. Droughts (e.g. Horn of Africa 2020–23) and floods destroy harvests; climate change is making weather less predictable.
Technology	Irrigation, fertilisers, pesticides, machinery and high-yield seeds increase production; many poor farmers cannot afford them.
Pests and disease	Locust swarms (East Africa, 2019–20) and crop diseases (e.g. cassava mosaic virus) can wipe out harvests; animal diseases kill livestock.
Water stress	Agriculture uses about 70% of global freshwater; shortages limit irrigation.
Conflict	War destroys farms, disrupts transport and drives farmers from their land — e.g. Sudan, Yemen. The war in Ukraine (2022) disrupted global grain exports and pushed up prices.
Poverty	Poor farmers cannot afford seeds, fertiliser or equipment, and poor families cannot afford to buy food even when it is available.

4. Impacts of Food Insecurity

- **Famine** — a widespread, severe shortage of food leading to starvation and death (e.g. Somalia 2011, when around 260,000 people died).
- **Undernutrition** — not eating enough of the right food, causing stunting (around 150 million children worldwide), weakened immune systems and diseases such as kwashiorkor.
- **Soil erosion** — desperate farmers overcultivate and overgraze land, removing vegetation and leading to soil erosion and desertification, which further reduces food supply.
- **Rising prices** — food price spikes (2008, 2022) push millions more into poverty.
- **Social unrest** — food shortages and high prices have sparked riots and contributed to political instability (e.g. the 2010–11 Arab Spring).

5. Strategies to Increase Food Supply

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Strategy	How it works	Evaluation
Irrigation	Artificially watering crops using canals, sprinklers or drip systems	Allows farming in dry areas and several harvests a year; can cause salinisation and depletes rivers and aquifers
Aeroponics and hydroponics	Growing plants without soil — roots in nutrient-rich water (hydroponics) or misted with nutrients in air (aeroponics)	Uses up to 90% less water, no pests from soil, can be done indoors in cities; expensive and energy-intensive
The new Green Revolution	Using biotechnology and sustainable methods — drought-resistant and pest-resistant crop varieties, better soil management — to raise yields, especially in Africa	The original Green Revolution (1960s) greatly increased yields in Asia but relied on expensive chemicals and benefited richer farmers most
Biotechnology / GM crops	Genetically modifying crops to increase yield, resist pests or add nutrients (e.g. Golden Rice with vitamin A)	Can boost yields and nutrition; concerns about effects on biodiversity and farmers' dependence on seed companies
Appropriate technology	Simple, low-cost technology suited to local needs and skills — e.g. treadle pumps, drip irrigation from recycled bottles, improved grain stores	Affordable, easy to maintain; small-scale, so limited impact on national food supply

6. Large-Scale Agricultural Development — Thanet Earth, Kent

Thanet Earth — the UK's largest greenhouse complex

- Opened in 2008 on the Isle of Thanet, Kent; the glasshouses cover around 90 hectares.
- It produces a large share of the UK's salad crops — hundreds of millions of tomatoes, cucumbers and peppers a year — using **hydroponics**, with nutrients delivered directly to the roots.

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- Rainwater is collected from the glasshouse roofs and stored in reservoirs; its own combined heat and power plants generate electricity, heat and CO_2 for the crops, with surplus electricity sold to the National Grid.

Advantages: year-round production; reduces food miles and imports; creates over 500 jobs; very efficient use of water and space.

Disadvantages: built on greenfield farmland; the gas-fired power plants produce emissions; light pollution at night; produce is grown for supermarkets, so local small farmers may struggle to compete.

7. Moving Towards Sustainable Food Supplies

- **Organic farming** — no artificial chemicals, protecting soils and wildlife; lower yields and higher prices.
- **Permaculture** — farming that copies natural ecosystems, e.g. mixing trees, crops and animals, so the system maintains its own soil fertility.
- **Urban farming initiatives** — growing food in cities on rooftops, allotments and vacant land, e.g. Incredible Edible Todmorden (West Yorkshire) or community gardens in Havana, Cuba.
- **Sustainable fish and meat** — avoiding overfished species (look for the Marine Stewardship Council label); free-range and lower-meat diets.
- **Seasonal food consumption** — eating food grown locally in season reduces energy used for transport and heating.
- **Reduced waste and losses** — about a third of food is wasted; better storage in LICs and less waste by consumers and supermarkets in HICs. In the UK, WRAP's campaigns ("Love Food Hate Waste") aim to cut household food waste.

8. Local Scheme in an LIC: The Makueni Food and Water Security Programme, Kenya

Makueni County, south-east Kenya

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- Makueni is a semi-arid area with unreliable rainfall and frequent droughts; many families depended on food aid.
- The UK charity **Excellent Development**, working with the Africa Sand Dam Foundation, helped local self-help groups to build **sand dams** across seasonal rivers. Sand builds up behind the dam and stores rainwater within it, protected from evaporation.
- Communities were trained in **terracing**, drought-resistant crops (sorghum, millet, cowpeas), tree planting and better seed storage.
- **Results:** families have water within easy walking distance all year; crop yields rose; households can grow vegetables and keep livestock, improving diets and incomes; time saved collecting water allows children (especially girls) to attend school.
- **Why it is sustainable:** it is low-cost, built by the community using local materials, needs little maintenance and does not rely on outside fuel or chemicals.

Exam Tip: A 9-mark question might ask "To what extent is sustainable food production possible?" Use evidence from local schemes (Makueni), urban farming and waste reduction, and weigh these against the scale of global demand and the role of large-scale agriculture such as Thanet Earth.

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